Operating System

Notes & Important Questions

Q1. Define Operating System and its types.

An **Operating System (OS)** is system software that acts as an interface between the user and computer hardware. It manages hardware resources and provides services to application programs.

Types of Operating System:

- **Batch Operating System** – Executes jobs in batches without user interaction.
- **Time-Sharing OS** – Allows multiple users to use the computer simultaneously.
- **Real-Time OS** – Gives output within a fixed time limit.
- **Distributed OS** – Manages a group of connected computers as a single system.
- **Network OS** – Provides services and manages resources over a network.
- **Multiprocessing OS** – Uses two or more processors to execute tasks.

Q2. What are the functions and services of OS?

Functions of Operating System:

- **Process Management** – Manages running processes.
- **Memory Management** – Allocates and manages main memory.
- **File Management** – Creates, deletes and organizes files and directories.
- **Device Management** – Controls input/output devices.
- **Security and Protection** – Protects data and system resources.
- **Resource Management** – Allocates CPU, memory and other resources.

Services of Operating System:

- Program execution
- I/O operations
- File-system manipulation
- Communication
- Error detection
- Resource allocation
- Protection and security

Q3. Define UNIX/Linux Architecture.

UNIX/Linux architecture is a layered structure in which different components work together to provide operating-system services.

Main components are:

- **1. Hardware** – Includes CPU, memory, hard disk, keyboard, etc.

- **2. Kernel** – The core of the operating system. It manages CPU, memory, files and devices.
- **3. Shell** – Acts as an interface between the user and the kernel. It accepts commands from the user.
- **4. System Utilities** – Programs used to perform various system-related tasks.
- **5. Application Programs** – Programs used by users for specific tasks.

Basic structure:

User → Applications → Shell → Kernel → Hardware

Q4. Define Services and System Calls.

Services:

Operating system services are the facilities provided by the OS to users and application programs.

Examples:

- Program execution
- File management
- I/O operations
- Communication
- Error detection
- Resource allocation

System Calls:

System calls are the interface through which a user program requests a service from the operating system kernel.

Examples:

- **open()** – opens a file
- **read()** – reads data
- **write()** – writes data
- **close()** – closes a file
- **fork()** – creates a new process
- **exec()** – executes a new program

Q5. Explain Operations and States of Operating System.

Operations of Operating System:

The OS performs various operations to manage the computer system, such as:

- Process management
- Memory management
- File management
- Device management
- Security and protection
- Resource allocation

Process States:

A process generally passes through the following states:

- **New** – Process is being created.
- **Ready** – Process is waiting for CPU allocation.
- **Running** – Process is currently being executed by CPU.
- **Waiting/Blocked** – Process is waiting for an I/O operation or some event.
- **Terminated** – Process has completed its execution.

Flow:

New → **Ready** → **Running** → **Waiting** → **Ready** → **Running** → **Terminated**